

Crop Alignment: A Structural-Null Test for Closed-Loop Labels in Segmentation Benchmarks Supplementary Material

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Paper ID *****

001	1. Threats to validity and evidence inventory		
002	The diagnostic has a narrow domain. Crop alignment	benchmark audit: an error list that only contains er-	034
003	is defined only on pixels for which overlapping crops	rors which turned out not to matter is not an error	035
004	receive different labels. If that set is empty, there	list.	036
005	are no observations from which to estimate κ . The	Two patterns account for most of them. The first is	037
006	method therefore cannot audit crop-invariant pseudo-	that a wrong number is more dangerous than a crash,	038
007	label shortcuts or shortcuts unrelated to crop geome-	because a crash stops the pipeline and a wrong number	039
008	try. It would be incorrect to turn an empty set into a	propagates. The second is narrower and worse: a check	040
009	zero and call it a negative result.	that reports success over a failure. Five of the entries	041
		below are of that kind, marked [silent]. None of the	042
		five raised, and all five would have passed unnoticed.	043
010	The contrast is not unique causal attribution. A	2.1. Errors that changed a number we had written	044
011	nonzero κ proves that a predictor changes with crop	down	045
012	context on the disagreement set. The matched control	1. The lead statistic was the wrong statistic. An ear-	046
013	removes the crop sensitivity learned under repaired la-	lier draft contrasted a spread of 0.1164 against seed	047
014	belts, but it cannot guarantee that the only remaining	variance. That figure is the unpaired standard devia-	048
015	treatment-control difference is imitation of the labeller.	tion of levels, and every comparison it was supporting	049
016	The experiments support a targeted diagnostic claim,	is paired, both arms on the same held-out acquisition,	050
017	not a mediation claim about a unique learning mecha-	so acquisition difficulty cancels. The correct framing	051
018	nism.	is a minimum detectable effect from the paired spread.	052
		The conclusion survived; the statistic did not.	053
019	The release has three evidence tiers. The primary	2. A fold exists only in a rounded text summary. The	054
020	sea-ice aggregate is recomputed from 16 paired values	committed JSON contains 16 aligned noisy/control	055
021	in committed JSON. Flood calibration and cost rows,	folds and recomputes to +0.1073 ($t = 12.88$). A	056
022	and the 60/120-epoch sea-ice results, are committed as	text summary contains a seventeenth row and reports	057
023	summary logs. Raw predictions, checkpoints, and some	+0.1106 ($t = 13.01$), but the corresponding per-fold	058
024	per-run JSON are absent. Thus the compact artifact	machine-readable output is absent. An earlier draft	059
025	verifies aggregate arithmetic but cannot regenerate ev-	promoted the text-only aggregate. The main paper	060
026	ery prediction or training run. A seventeenth rounded	now uses only the 16-fold result and discloses the gap;	061
027	sea-ice row without matching JSON is excluded from	a larger number is not evidence when the release can-	062
028	the primary claim.	not reconstruct it.	063
029	2. Errors in our own analysis		
030	The main paper discloses the artifact-provenance fail-	3. Significance computed after averaging away the vari-	064
031	ure that changes its headline sample. This appendix	ance. Gate 3 averaged confusion matrices across seeds	065
032	records that issue and the rest of the analysis audit.	and then bootstrapped, which removes seed variance	066
033	We report them for the same reason we report the	from the interval. Recomputed as paired differences	067

068	across the five seeds, none of the three contrasts is significant.	2.2. Checks that reported success over a failure	113
069	Two claims stated as significant were withdrawn (Section 3).	10. A convergence test nothing could satisfy [silent].	114
070		epoch_control.py asked whether best_epoch $\geq$ 60	115
071	4. A $7.2\times$ contradiction between adjacent sections.	for runs capped at 60 epochs. No run can satisfy that,	116
072	The deficit at two acquisitions appeared as -0.0295	so the script printed “every long run converged before	117
073	from the controlled learning curve and -0.2121 from	the cap” over best epochs of 59, 56 and 51. The cri-	118
074	a different table. They are not two estimates of one	terion that carries meaning is whether early stopping	119
075	quantity: they differ in data, in scoring convention,	ever fired.	120
076	and in training budget. Presented on one axis they		
077	implied a disagreement that did not exist.	11. An octal literal that killed 33 of 55 jobs	121
078	5. Three spreads for one quantity. The 12-epoch pre-	[silent]. A sweep derived its parameter with python	122
079	mium is recorded with $s = 0.0859$ in one runner header,	-c "print(\${A}/100.0)" on a zero-padded tag. Python	123
080	0.0910 in the paper, and an implied 0.1003 behind a	rejects 050 as a malformed octal literal, while 000 and	124
081	required- n of 238 quoted in the same header. The cur-	100 are both legal. Exactly the two ends of the sweep	125
082	rent computation gives 0.0910 and 197 reproducibly.	survived and all 33 interior jobs died, and the loop	126
083	The header predates a fix and was never recomputed.	announced completion with 22 of 55 results on disk.	127
084	6. A hand-picked subset described as balanced. The	Running the sweep ends-first is the only reason this	128
085	runner for the six-fold budget preview records its folds	cost jobs rather than the claim.	129
086	as “the two largest, the two smallest, and the one neg-		
087	ative fold.” The benchmark has three negative folds.	12. A type coercion that ate an underscore [silent].	130
088	The preview included one, and excluded the largest	pd.read_json coerced acquisition identifiers to int64,	131
089	fold, both of the two smallest positives, and the ac-	so a drift test matched nothing, reported “0 complete	132
090	quisition that later turns out to move most between	folds” and exited 0.	133
091	training budgets. We had recorded the preview’s fail-		
092	ure as an error of subset size. It was a selection rule	13. A layout check that could not match its own pat-	134
093	that was never checked against the data it claimed to	tern [silent]. Overfull lines were counted with a shell	135
094	describe.	command in which the doubled backslash was literal,	136
095	7. Bonferroni tolerance overstated. A claim that	so it reported zero while three tables overflowed their	137
096	every headline result survives a Bonferroni factor of	column. The error was found from a rendered page	138
097	100 was true for two of three. The accuracy cost at	image, not from the log.	139
098	$t = -4.27$ is $p = 0.0016$ on 10 degrees of freedom and		
099	tolerates a factor of about 30. The main paper now	14. A negative control that tested nothing [silent]. To	140
100	states the weakest case rather than the strongest.	confirm the layout check could detect a fault, a delib-	141
101	8. A coefficient tested against the wrong reference.	erate break was introduced with a substitution whose	142
102	The threshold-arm coefficient β was tested against 0,	pattern never matched. Nothing was broken, the check	143
103	which it obviously differs from. The meaningful refer-	passed, and the pass was recorded as evidence that the	144
104	ence is $\beta = 1$, the value the difference in differences	check worked.	145
105	assumes. Against that reference $\hat{\beta} = 0.711$ is not sig-	2.3. Errors that a build or a log would not report	146
106	nificant, which is the honest reading and the one the	15–16. Two control characters from shell quoting.	147
107	paper reports.	\times became a tab, rendering “2.7imes” in the com-	148
108	9. Independence claimed for nested settings. Five dial	piled PDF; \approx became a BEL byte. Neither pro-	149
109	settings were described as five independent tests. The	duced a build error and both exit 0. Both were in-	150
110	dial nests them by construction and they share the	troduced by escaping a LaTeX macro through a shell	151
111	same eleven events, so what they show is consistency	string, and both were found by scanning the source	152
112	across a sweep, not five separate results.	bytes rather than reading the log.	153
		17. A figure that contradicted its own text. The main	154
		text declines to quote a required sample size for crop	155
		alignment, because the power formula returns a value	156
		below two and a paired test cannot estimate a variance	157

158 from one fold. The figure drew that same quantity as
159 a bar labelled “1”.

160 18. Four floats never referenced. LaTeX does not
161 warn about a table or figure that no text points at.
162 Four were unreferenced, which also degraded their
163 placement.

164 19–20. A renumbering trap, twice. A cross-reference
165 to a section that did not yet exist has no heading to
166 anchor against, so a renumbering pass moved it to a
167 different section. The dangling-reference check passed
168 both times, because the wrong target exists.

169 3. Claims withdrawn from an earlier version

170 Gate 3 significance. “Every difference is significant
171 under the paired bootstrap” and “the true-photon arm
172 is significantly worse than a constant” were both stated
173 in an earlier version. Recomputed at the run level
174 across five seeds, neither holds. Withdrawn.

175 The seed-variance framing. An earlier version argued
176 that acquisition spread exceeded seed variance using
177 an unpaired standard deviation of levels. Recomputed
178 as a paired quantity, seed variance is 0.31% of the pre-
179 mium variance, which is a stronger statement than the
180 one withdrawn, but the original argument was not ev-
181 idence for it.

182 “The leak is eliminated.” Three nulls were accepted
183 at $n = 3$ with a spread near 0.07, where an effect
184 of 0.03 would have been undetectable. “Eliminated”
185 overstated “not detected”. Because the release contains
186 no independently justified equivalence margin, the re-
187 vised paper treats these as underpowered absence-of-
188 evidence statements and does not use them to claim
189 elimination.

190 A prediction we made and lost. We derived a point
191 prediction for the patch-versus-mosaic gap with noth-
192 ing fitted. The intercept lands where theory puts it
193 and the sign agrees on 14 of 17 folds, but the magni-
194 tude is out by a factor of 1.6 to 1.9. The prediction is
195 reported in the main paper as made and refused, rather
196 than removed.